CLASSROOM MANAGEMENT SYSTEM

File-Based Version

1. Project Title

Classroom Management System — File-Based Version

2. Group Number: Group 1

3. Group Members

1. Mohammed Abu — Representative
2. Abdulmenan
3. Abdulazeez Rashid
4. Barento Siraj
5. Hayder Abdu
6. Bontu Abuti

4. Project Objective

The objective of this project is to develop a simple Python-based Classroom Management System for managing student academic records. The system helps users add, search, update, delete, and display student information while automatically calculating averages, grades, and Pass/Fail status.

5. Main Features

The system provides the following features:

- Add new student records
- Validate student scores between 0 and 100
- Calculate student averages automatically
- Assign grades based on average scores
- Determine Pass or Fail status
- Search students by ID
- Update student information
- Delete student records
- Display all student records
- Display class performance summary
- Save and load student records using a text file
- Save records before exiting the program

6. Data Structures Used

The project uses:

- List: Stores multiple student records.
- Dictionary: Stores individual student information such as ID, name, department, scores, average, grade, and status.
- Set: Stores student IDs and helps prevent duplicate IDs.

7. Functions Used

The program is organized into functions such as:

- "calculate_average()"
- "calculate_grade()"
- "calculate_status()"
- "get_score()"
- "add_student()"
- "display_all_students()"
- "search_student()"
- "update_student()"
- "delete_student()"
- "class_summary()"
- "save_to_file()"
- "load_from_file()"
- "exit_program()"
- "main_menu()"

These functions make the program organized, reusable, and easier to maintain.

8. File Handling

Student records are handled using the text file "group_1_students.txt". The "save_to_file()" function writes student information to the file, while "load_from_file()" reads records from the file. This allows student information to be stored outside the running Python program.

9. How the System Works

When the program starts, it displays a main menu with different options for managing students. When adding a student, the system the

collect student's ID, name, department, and three scores. It validate the scores, calculates the average, assigns a grade, and determines the student's status.

The student information is then stored in the program using a dictionary and list. Users can search, update, delete, or display records. The class summary provides an overview of student performance. The system also provides options to save and load student records.

10. Individual Contributions

Mohammed Abu: Coordinated the project and contributed to system design, coding, integration, debugging, testing, and documentation.

Abdulmenan: Contributed to student record management, testing, and debugging.

Abdulazeez Rashid: Contributed to student input, score validation, grading, and testing.

Barento Siraj: Contributed to search, display, and update features and system testing.

Hayder Abdu: Contributed to file handling, record management, testing, and documentation.

Bontu Abuti: Contributed to testing, documentation, and project review.

11. Conclusion

The Classroom Management System demonstrates the practical use of Python programming concepts, including functions, lists, dictionaries, sets, loops, conditional statements, input validation, and file handling. The project provides a simple solution for managing student academic records and demonstrates the group's teamwork and programming skills.